

SAKETH REDDY

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EDUCATION

Georgia Institute of Technology

May 2026

Major: B.S. Computer Science, Specializations in System Architecture and Cybersecurity

GPA: 3.95

Coursework: Systems for Machine Learning, Operating Systems, Compilers, Honors Algorithms

Special Topics Binary Exploitation, Honors Discrete Math, Data Structures and Algorithms, Linear Algebra

Combinatorics, Privacy Technology Policy and Law, Enterprise Cybersecurity Risk Management

TECHNICAL SKILLS AND AWARDS

Languages: Python, C, Go (Golang), Rust, C++, Java, Bash, x86 Assembly, SQL, KQL, HTML, CSS, Javascript, React

Libraries, Tools: Docker, Git, GDB, RegEx, ADX, LLMs, OpenAI API, angr, z3, AFL++, Flask, Ghidra, pwntools

Awards: DARPA AIXCC Finalist, 2nd Place DEFCON CTF Finals, 1st Place AI ATL Hackathon, National Cyber Scholar

EXPERIENCE

Software Engineer Intern (Security Research R&D) | Go, Python, KQL, SQL

May 2025 – Jul 2025

Microsoft

Redmond, WA

- Used Go to implement Linux persistence mechanism extraction and analysis feature leveraging the Defender antivirus engine shared library to produce malware reports of programs identified as autostart execution points
- Implemented 9 Domain Specific Language parsers with corresponding fuzzers covering all untrusted data sources
- Achieved 85.9% test coverage of feature by leveraging Dependency Injection to create 159 unit & integration tests
- Found critical security bug in Microsoft Defender antivirus engine. Bug enabled malware to evade detection and cause system instability. Facilitated cross-team handoff to feature owners. Issue escalated to CISO-level visibility
- Won 1st place in security impact category at organization-wide final intern showcase

Research Assistant | Python, Rust, C, Docker, LibAFL, Fuzzing, Git

Mar 2024 – Present

SSLab @ Georgia Institute of Technology | Advisor: Dr. Taesoo Kim

Atlanta, GA

- Won \$2 Million with Team Atlanta in the DARPA AI Cyber Challenge by developing Atlantis, a LLM-based Cyber Reasoning System to autonomously discover and patch zero-day vulnerabilities. Found zero-day vuln in SQLite3
- Created Python-in-Rust fuzzing mutator adapter for LibAFL using PyO3, enabling LLM-generated mutator functions
- Developed semantic normalization and equivalence algorithms for Testlang DSL, a Backus-Naur form Domain Specific Language enabling LLM agents to understand fuzzing harnesses by characterizing their input formats

Security Research Intern | Python, Rust, Docker, LangChain, LibAFL, GDB, PwnTools

Jun 2024 – Aug 2024

Air Force Research Lab | Advisor: Dr. Matthew Anderson

Rome, NY

- Developed LLM agent for autonomous program analysis, solving 84% of relevant CSAW CTF Qualifier challenges from NYU CTF dataset in <3 minutes by leveraging custom dockerized Bash, Python, and GDB agent tooling
- Created LLM-based directed fuzzing mutator using Rust, Python, LibAFL, and Claude 3.5 Sonnet, outperforming sota AFLGo in directed function coverage efficiency by leveraging semantic understanding of input grammar

LEADERSHIP

President, CTF Competition Captain

Mar 2024 – Present

GreyHat Cybersecurity Club @ Georgia Institute of Technology

Atlanta, GA

- Led 24 CTF training workshops focusing on Reverse Engineering, Web Exploitation, and Binary Exploitation
- Grew club membership by 140% over the course of 1 year by restructuring to prioritize 1:1 mentorship
- Led team to win 2nd place at international-level DiceCTF Finals Undergrad Jeopardy category in NYC

PROJECTS

Last-N Context LLM Agent for C Vulnerability Patching | Python, Docker, C, PinTool, GDB

January 2025 – Present

- Developed LLM-based vulnerability patching engine centered around reasoning LLM models and function call recency heuristic, patching 90.4% of nginx AIXCC semifinals challenges and ~70% of sampled OSSFuzz C vulns
- Used Python, Intel PinTool, Docker, and gdbscript to instrument and dynamically extract the last n=30 unique call instruction targets from OSS-Fuzz crash traces in order to generate function recency heuristic
- Achieved 32x speedup by leveraging multithreading and multiprocessing to parallelize end to end evaluation framework, enabling evaluation of novel branched assertive function patch strategy
- Engineered Dockerized e2e patch framework benchmark, enabling batch processing of containerized experiments